

HW3 Symbolic Music Generation

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Code&Results: https://drive.google.com/drive/folders/1M_rN-MuIWTwl49ijwWST31OzzTykVJ9i?usp=drive_link

Outline

- Task1 Unconditional Generation
 - Token Representation and Model Architecture
 - Workflow
 - Implementation Detail
 - Inference Configurations
 - Results
- Task2 Continuation Generation
 - Token Representation and Model Architecture
 - Workflow
 - Inference Configurations
- Conclusion

Outline

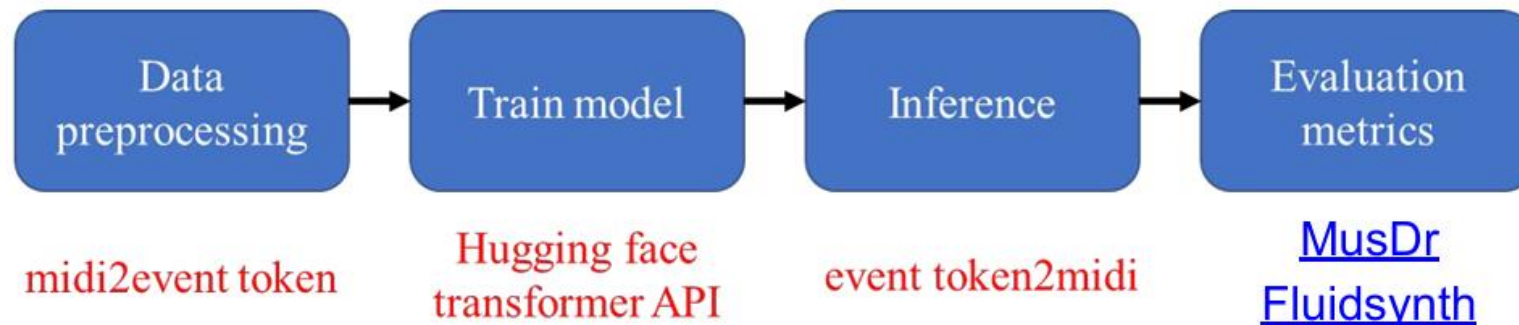
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Token Representation and Model Architecture

- Token Representation
 - Adopt the REMI representation
- Model architecture
 - The backbone model is GPT-2
 - Uses the default configuration from GPT2Config
 - An autoregressive Transformer for symbolic music generation

Workflow

1. Data Preprocessing
 - Convert MIDI files into event-based tokens (midi2event token)
2. Model Training
 - Train the GPT-2 model using the HuggingFace Transformers API
3. Inference
 - Convert generated event tokens back to MIDI (event token2midi)
4. Evaluation Metrics
 - Use MusDr and FluidSynth to synthesize audio and evaluate musical quality



Implementation Detail

- Sequence Length (X_LEN): 1024
- Event Dictionary: basic_event_dictionary.pkl
- Epochs: 100
- Batch Size: 4
- Learning Rate: 0.0002

Implementation Detail (Chord Extraction)

- Integrated chord extraction based on the official REMI GitHub implementation
- Imported the chord dictionary (chord_event_dictionary.pkl) including pitch classes and chord maps
- Updated the event sequence construction from
items = tempo_items + note_items
to
items = tempo_items + note_items + chord_items

Inference Configurations

- Temperature: 1.2
- Top-k: 5
- Since autoregressive generation continuously expands the sequence, the inference process restricts the input length to the first 1024 tokens for next-token prediction
- The submitted inference results are generated using the Epoch-50 checkpoint

Results

Id	Model	Representation	Event	Loss (epoch)	Top-k	Temperature	H1	H4	GS
1	GPT2	REMI	w/o chord	0.6549 (25)	5	1.2	1.8101	2.4007	0.8630
2	GPT2	REMI	w/o chord	0.2551 (50)	5	1.2	2.0307	2.5175	0.8207
3	GPT2	REMI	w/o chord	0.1094 (100)	5	1.2	1.9174	2.4641	0.8639
4	GPT2	REMI	w/o chord	0.1094 (100)	3	1.2	2.0806	2.5571	0.8139
5	GPT2	REMI	w/o chord	0.1094 (100)	10	1.2	1.9082	2.4921	0.8805
6	GPT2	REMI	w/ chord	0.2270 (50)	5	1.2	1.5497	2.1447	0.8369
7	Real data						1.9355	2.5273	0.7790

* **Id2** was selected as the final submission

Results

- ID2 (Epoch 50, no chord, top-k=5) delivers the best overall balance => chosen as final submission
- Longer training improves loss but not always musical metrics
- Top-k = 5 is the most stable setting
- Chord extraction does not improve generation under the current model size
- Generated music is close to real data in H1, H4 and GS

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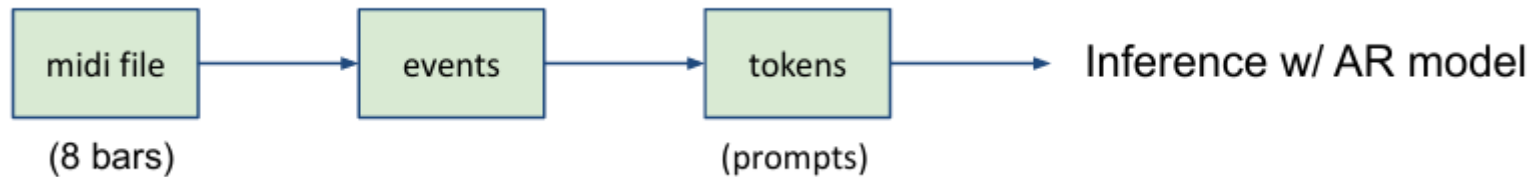
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Workflow

1. Input MIDI file (8 bars)
2. Convert to event sequence
3. Transform events into tokens (prompts)
4. Perform symbolic music generation using the autoregressive (AR) model



Inference configurations

Config	Epoch	Top-k	Temperature	Chord Extraction
Config1(task1 id2)	50	5	1.2	Without chord
Config2 (task1 id3)	100	5	1.2	Without chord
Config3 (task1 id6)	50	5	1.2	With chord

*Config2 produced the best generation quality among all settings, so **Config2** was selected as the final submission.

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Conclusion

- This project successfully implemented GPT-2 with the REMI representation for both Task 1 (Unconditional Generation) and Task 2 (Continuation Generation).
- After comparing multiple inference settings and chord-extraction variants, **Task 1 id2** achieved the best balance among H1, H4, and GS, and was selected as the final submission. For **Task 2**, the **id3** produced the best continuation quality and was therefore chosen as the best generation.
- Although chord extraction introduces additional harmonic information, it did not outperform the non-chord configurations under the current model size and training setup.
- Overall, the model is capable of generating musically coherent symbolic sequences, demonstrating the effectiveness of GPT-2 for symbolic music modeling.

Reference

1. <https://github.com/YatingMusic/remi>
2. <https://github.com/slSeanWU/MusDr>
3. HW3.pdf